A SQL PORTFOLIO PROJECT

Mortgage lending in New York

Every mortgage application filed in New York State over four years, as the lenders reported it. Modelled in PostgreSQL, analysed in forty-two queries, and runnable in your browser without installing anything.



1,596,676

Applications decided

951,371

Loans originated

$371.4bn

Disbursed

Samuel Babajide · HMDA, New York State, 2022 to 2025

swipe →

THE MARKET

Four years, year on year

YEAR	APPLICATIONS	ORIGINATED	DISBURSED	APPROVAL	DENIAL
2022	500,553	307,252	$128.5bn	64.39%	18.69%
2023	346,943	203,072	$72.7bn	61.72%	21.68%
2024	355,249	208,814	$76.9bn	61.94%	21.02%
2025	393,931	232,233	$93.3bn	62.71%	19.63%
All four	**1,596,676**	**951,371**	**$371.4bn**	**62.85%**	**20.09%**

Applications fell 30.7 percent from 2022 to 2023 and have recovered 13.5 percent since. Approval bottomed in 2023 at 61.72 percent and denial peaked the same year at 21.68. 2022 is a three-year file and 2025 is provisional, so the four vintages are not equally complete.

THE TYPICAL LOAN

What a New York mortgage actually looks like

$127,000 MEDIAN APPLICANT INCOME	**$475,000** MEDIAN PROPERTY VALUE	**$245,000** MEDIAN LOAN AMOUNT
75.00% MEDIAN LOAN TO VALUE	**1.96×** MEDIAN LOAN TO INCOME	**6.281%** MEDIAN INTEREST RATE

Across the 951,371 loans that were actually made. Every median uses **percentile_disc**, which returns a value some borrower really had, rather than interpolating between the two middle rows and reporting a figure nobody was charged. That choice is also what makes these numbers identical in PostgreSQL and in the browser.

COVERAGE

What the register covers

4	1,168	62	5,281
YEARS	LENDERS	OF 62 COUNTIES	CENSUS TRACTS

4	6	4	9
LOAN TYPES	LOAN PURPOSES	SEX VALUES	RACE VALUES

MOST EXPENSIVE COUNTY	MEDIAN VALUE	LOANS
New York County	$1,355,000	35,463
Kings County	$1,165,000	53,975
Queens County	$805,000	57,169

All 62 New York counties appear. The raw column returns 66, because it also carries an **UNKNOWN** value and three out-of-state codes filed against New York property. Counting those would have claimed four counties that do not exist.

APPLICANT DEMOGRAPHICS

Originated loans by applicant race

RACE	SHARE OF ORIGINATIONS	SHARE	DISBURSED
White 621,461 loans		**65.32%**	$198.2bn
Race not available 161,122 loans		**16.94%**	$98.9bn
Asian 87,024 loans		**9.15%**	$44.1bn
Black or African American 58,130 loans		**6.11%**	$19.6bn
Joint 17,886 loans		**1.88%**	$8.9bn
Four smaller categories 5,748 loans		**0.60%**	$1.7bn

The regulator's categories are never dropped here, including "not available". It is 16.94 percent of originations and 26.6 percent of the dollars, so removing it would move every other share on this slide without saying so. The six rows sum to 951,371 loans exactly.

APPLICANT DEMOGRAPHICS

Originated loans by applicant sex

SEX	SHARE OF ORIGINATIONS	SHARE	DISBURSED
Joint 338,370 loans		**35.57%**	$130.9bn
Male 312,145 loans		**32.81%**	$111.9bn
Female 210,493 loans		**22.13%**	$59.6bn
Sex not available 90,363 loans		**9.50%**	$69.0bn

Look at the last row twice. "Sex not available" is 9.50 percent of the loans and **18.57 percent of the dollars**, the only row where the two diverge sharply, on a median loan of **$335,000** against $265,000 for joint applicants. These are largely applications filed by a company rather than a household, which is why the field is blank, and it is the clearest reason not to delete a "missing" category.

DENIAL ANALYSIS

The three reasons lenders write down most

REASON CITED	SHARE OF ALL CITATIONS	SHARE	CITATIONS
Debt-to-income ratio		32.87%	135,207
Credit history		21.71%	89,302
Collateral		17.45%	71,778

The unit is a citation, not an application. HMDA lets a lender give up to four reasons, so 320,762 denied applications carry **411,362 citations**. These three are **72.03 percent** of them and seven more reasons share the rest. Debt-to-income alone is cited half as often again as credit history.

WHAT THE DATA RECOMMENDS

Three findings a borrower can act on

1 Lower is not better, and the reason matters

Under 20 percent debt-to-income denies at **24.49 percent**, worse than every band from 20 to 49, where the rate is flat near **13**. That bottom is not cautious borrowers: it is small second-lien business. The threshold is a line, not a direction. Get under 50. Below 20 buys nothing.

2 Which lender you ask is worth 15 times

Same product in every row, conventional first-lien home purchase, 72 lenders with 2,000 or more such applications. Denial runs **1.14 to 31.57 percent**. SEFCU denies **1.14** on **more leveraged, lower-income** applicants than Bank of America, which denies **17.76**. Most borrowers approach one lender.

3 Borrowing against a home you own is a different market

Home improvement denies at **35.13 percent** and other purpose at **35.84**, against **10.78** for a purchase. With **493 and 479** lenders active rather than 934. Three times as hard, with half the choice.

THE MODEL

99 flat columns into 17 tables

1 PARTITIONED FACT	**4** YEARLY PARTITIONS	**8** DIMENSIONS	**2** OF THEM JUNK
4 BRIDGE TABLES	**1** QUARANTINE TABLE	**3** REFERENCE TABLES	**3** LABELLED VIEWS

md5 business key · year-keyed conformance · exact decimal arithmetic · 42 queries, two engines

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175 columns, 100 percent documented. The published data dictionary is generated from **pg_catalog** at build time, so it cannot describe a column that no longer exists.

HMDA strips every loan identifier before publication, so the business key is an **md5 of the source row**. Rebuild the model on another machine and you get the same keys.

DON'T TAKE MY WORD FOR IT

Run the queries yourself

42

queries, each one printed beside its result

2

engines, agreeing cell by cell

0

installs, accounts or servers

1.75 million rows of Parquet, queried in your own browser tab by DuckDB compiled to WebAssembly. Nothing is installed, no account exists and no data leaves your machine. Every figure in this carousel is one click from the query that produced it, and an automated parity test runs all forty-two against PostgreSQL 18 and DuckDB and fails on any difference. **Forty-two of forty-two agree.**

→ The playground

→ The data dictionary

→ The ERD

→ The repository

samuelbabajide.github.io/us-home-mortgage-disclosure-act-new-york

YOUR TURN

What would you ask this data?

Open the playground, change a query, press **Ctrl + Enter**. If you find something I missed, I would like to know.

BUILT WITH

PostgreSQL 18 · dimensional modelling · data quality assertions · query optimisation · Python · DuckDB WASM · Parquet · GitHub Pages

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